

Management of maize streak virus

Recognize the problem

Maize streak virus is very common in Ethiopia and is found in almost all maize growing areas. It has been reported to cause disease incidences that vary from a few infected plants per field to total yield loss with 100% infection. Erratic epidemics have been occurring every 3-10 years. The main damage caused is to plants younger than six weeks old. On young plants, the top and bottom surfaces of leaves have yellowish and light green streaks while mature plants have whitish, yellow and light green streaks running along their surface in the direction of the leaf veins. Maize plants can be severely stunted if the crop is attacked during the 4-5 leaf stage. Abnormal bunching of flowers and shoots and reddish pigmentation may also be seen.

Symptoms of maize streak virus on the leaves. (Photo by CIMMYT, flickr.com)



Background

Maize streak virus disease is caused by a virus, transmitted by a brownish white leafhopper that feeds on the maize leaves. It is highly persistent in the vector. The more leafhoppers are in the field, the quicker the disease will spread.

The vector which spreads maize streak virus - leafhopper. (Photo by S. Rae, flickr.com.)



Management

- Use certified disease free seed from a registered stockist and plant at the onset of the rains
- Planting a large area of maize all at once is likely to make the crop less vulnerable to maize leafhopper infestation
- Inspect the field regularly when the maize is small, looking for diseased plants
- Uproot infected plants when they first show signs of disease. This will keep the disease from spreading to healthy plants. Put the whole plant in a sack so the leafhoppers do not move to other plants
- Remove infected maize plants (rogueing) at an early stage and feed the removed plants to animals
- Keep the fields free from weeds, in particular grasses, to keep the vectors away and reduce disease transmission
- Remove cereal crop residues since they serve as an infection source
- Use chemical insecticides eg. Gaucho 70 WS (imidacloprid) 125gm/ha for the control of the vector before virus infection

When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) > **Maize streak virus**

The recommendations in this factsheet are relevant to: Ethiopia



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